Trenitalia
ETR1000 VZI R1

Air Supply

Revision	Author	Date	Revision Description
00.1	R. Schiavo	27/06/2023	Draft Release Platform.-.Zefiro-Europe/2F_06.-.Auxiliaries-VF.-.Zefiro-Europe/2F_06.03.-.Air Supply-VF.-.Zefiro-Europe/Air Supply-VF.CONCEPT, current 4.0 doc 3ECP41379-603 reviewed for VZI-6
01	G.Tarquini	22/11/2023	Additional compressor modes: Parking mode Fault reaction for compressor continuous running (additional switches of the master to slave compressor) Modified: 3.5 Air Dryer (cycling time is 120 s and not 60s) 2F_06.03.Zefiro-Europe.CONCEPT.133 (cycling time is 120 s and not 60s) 3.6.1.1 Normal Condition
01.01	G.Tarquini	14/02/2024	Changed project name according to ANSFISA re-naming Open Item clarification from email [VZI-R1] Dubbi FRS Air Supply 2F_05.02.Zefiro-Europe.CONCEPT.HRAS002 cancelled 2F_05.02.Zefiro-Europe.CONCEPT.HRAS003 modified: compressor swap is limited to compressor overheating or continuous running 2F_05.02.Zefiro-Europe.CONCEPT.HRAS002 clarified how to track the information 2F_05.02.Zefiro-Europe.CONCEPT.HRAS005 added information for the timer counter for compressor swap
01.02	G.Tarquini	13/04/2024	Added additional clarifications for 2F_05.02.Zefiro-Europe.CONCEPT.HRAS003, 2F_05.02.Zefiro-Europe.CONCEPT.HRAS001 as consequence of SW development questions. No changes to the content of these requirements.

Content

1 Introduction 4

1.1 Train Configuration 4

1.2 Definitions and abbreviations 4

1.1.1 Requirement ID 4

2 System Overview 5

2.1 System Configuration 5

3 Functional requirements 6

3.1 Control of Main Compressor 6

3.1.1 Function Design 6

3.1.2 Requirements 6

3.2 Monitoring of Main Compressor 13

3.2.1 Function Design 13

3.2.2 Requirements 13

3.3 Forcing of Main compressor 15

3.4 Monitoring of Main Pipe 16

3.4.1 Function Design 16

3.4.2 Requirements 16

3.5 Air Dryer 18

3.5.1 Function Design 18

3.5.2 Requirements 18

3.6 External Air Supply 19

3.6.1 Function Design 19

3.6.2 Requirements 19

3.7 Control of Auxiliary Compressor 20

3.7.1 Function Design 20

3.7.2 Requirements 21

3.8 Suspension Supervision 23

3.8.1 Function Design 23

3.8.2 Requirements 23

3.9 Provision for Tilt system 24

3.9.1 Function Design 24

3.9.2 Requirements 24

4 Signal Interface 26

4.1 I/O Interface 26

4.1.1 Supplier system 26

4.1.2 TCMS SW 26

4.2 Bus Interface 27

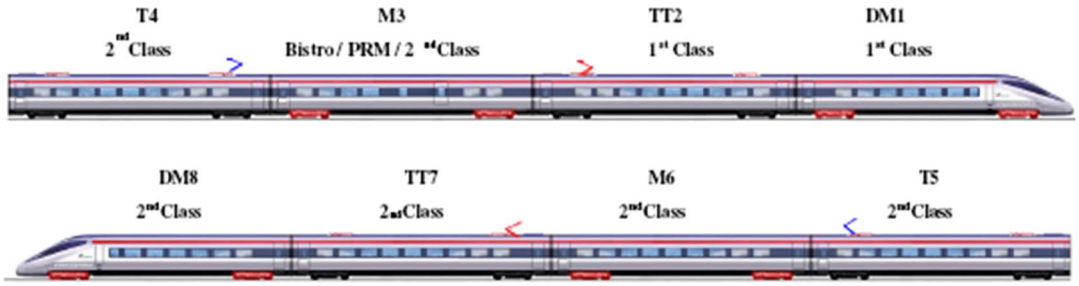
5 References 28

1 Introduction

This document is a function specification for the control of the air supply system for ETR1000 VZI R1 for customer Trenitalia.

1.1 Train Configuration

Each ETR1000 VZI R1 train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Intelligent Display Unit
TCMS	Train Control and Management System
MR	Main Reservoir
MRP	Main Reservoir Pressure
EB	Emergency Brake

1.1.1 Requirement ID

First counter (ID) requirement to be used for additional requirements: 2F_05.02.Zefiro-Europe.CONCEPT.HRAS006

2 System Overview

2.1 System Configuration

Air supply system includes 2 air supply units at 8 car train. Each unit includes main compressor and auxiliary compressor. Main compressor is oil-free double-stage piston type reciprocating compressor. The compression is performed in a double stage configuration in order to produce an intercooled compression principle from ambient to discharge pressure.
The following components are referred to in the document:

Equipment in DM cars:
TCP 13.12: Pressure sensor in MR pipe, located in brake panel 04.13.
PCP: Pressure switch MRP 650/750kPa(open/close)
RIHVAC 20.03: Isolation cock with micro switch indicating HVAC is isolated from MR.

Equipment in T4/5 car:
TCC 13.23: Pressure sensor in MR pipe for main compressor control, located in air treatment unit 13.03 in T-car
PSP 08.09: Pressure switch for auxiliary compressor, located in air compressor module 08.01.
PTP/PTA (part of 13.05): Pressure switches to supervise air dryer towers switching.
RIPO 13.29: Isolation cock with micro switch indicating external air supply is connected.
RISP 13.22: Isolation cock with micro switch indicating air supply is isolated from MR.

All cars:
RISOSP: Isolation cock with micro switch per car for secondary suspension.
The main compressor control for ETR1000 is performed on a 8-car train with 2 working compressors. Compressors are working on a start/stop basis. The main compressor has higher start up pressure level than the start up pressure for secondary compressor. At first start-up of train both compressors are run simultaneously to achieve pressurised train at shortest possible time. During normal operation only the main compressor is in operation but if the pressure drops below the start up level on the secondary compressor even that compressor is started. Both compressors shall stop at the same pressure level measured on the main pipe.
In case of connected external air supply main compressors are blocked.
When operating the compressors the following three main criteria are important:

1. quick start-up from empty train
2. equal usage of compressors
3. optimised operational load cycle

The compressor control will be done in such a way that ensures equal running time for both compressors.
To prevent the compressor not to start against back pressure there is a pressure switch that monitors the pressure and after a stop blocks the compressor until the pressure within the compressor house has decreased below an acceptable level. In addition the compressor is supervised and protected against overheating by a temperature switch blocking the compressor when temperature in the motor windings is too high.
The compressor is provided with an input air filter providing the proper filtration for the dusty climate.
There are two types of compressors in 8-car ETR1000. The main compressors (2) supply pressurized air to the consumers, as brakes, secondary suspension, pantograph, semaphores, toilets, HVAC, horn, flange lubrication and etc. They are placed in T4- and T5-car. Main compressors are dependent on auxiliary 3-phase 400 VAC supply.
In the same air supply unit in T4- and T5-car there is also an auxiliary compressor running on 110VDC, which is used to raise the pantograph during start up.

3 Functional requirements

3.1 Control of Main Compressor

3.1.1 Function Design

Compressors motor is switched on via 3 phase contactor available on the train and managed from train control. If the pneumatic system pressure falls down below 850 kPa one compressor starts. When pressure achieves 1000 kPa the compressor is switched off.
Air treatment unit operation is controlled with the air dryer control unit which manages dryer valves, draining valves and heating system. The heaters are activated when temperature falls down below +5° C. Second thermal switch is set to -25° C. Below this temperature the dryer does not operate for 10 minutes after that dryers are working normally.
An internal thermal switch alerts over temperature for the main electric motor.

3.1.1.1 *Normal Condition*

Normal condition is where primary compressor works between 850 -1000 kPa and secondary compressor works between 800-1000 kPa. Compressors are controlled by pressure transducer, connected to BCU, placed in each air supply unit. Main reservoir pressure is normally between 800-1000 kPa when train is activated and start up is done.

All main compressors are running simultaneously during start up if 3~400V AC is available until train is pressurized (1000kPa). Primary and secondary compressor is defined to optimise the running time. In normal operation the primary compressor will be running while the secondary compressor acts as a backup. Run time is supervised, optimised and saved as condition data in non volatile memory.

3.1.1.2 *Failure Cases*

1. Only one compressor starts
2. No compressor starts (double fault)
3. High leakage in MR system
4. Pressure sensor has failed in one compressor control panel

3.1.1.3 *Failure Detection*

1. An event will be generated
2. An event will be generated
3. Low main reservoir pressure, emergency brake not released.
4. An event shall be generated

3.1.1.4 *Failure Handling*

1. No feedback from contactor for one compressors. Indication on IDU which compressor has started. One compressor is enough to fulfil the need of compressed air in the train. No effect on train operation during normal compressor management, at start-up filling time will be increased.
2. No filling of the reservoirs, not maintaining the pressure. Stopping error, not possible to drive the train.
3. No pressure increase in main reservoir. Indication on IDU and pressure gauge in drivers cab. Stopping error if the main compressors cannot maintain reservoir pressure. The Leakage shall be blocked before train is taken into service.
4. Both compressors shall be controlled by the other non faulty sensors. No effect on train operation.
5. In case of both compressors have been disabled due to the failure at least of one air dryer, the driver is able to re-activate the not faulty compressors for a maximum time to finish the daily service, still ensuring an acceptable air quality for the pneumatic system. Once the vehicle is in the depot, failures must be repaired and the maintainer shall be reset the bypass.

3.1.2 Requirements

The following pressure sensors are used for compressor control:
Compressor control pressure sensor is TCC in T4/T5.
MR -pressure sensor TCP in DM1/DM8

Description	All main compressors shall run during start up of the train when the main reservoirs are empty and when 3~ 400V AC supply is available. The shortest possible start up time shall be ensured. After achieving normal operational conditions the run time of the compressors shall be optimised.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.46	All main compressors shall be ordered to run during start up of the train when MR pressure is less than 800 kPa, after 3□400V AC is available.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.415	After first start and MR pressure has reached 1000 kPa compressor management shall be done according normal operation between 850 and 1000 kPa.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.470	When the main compressor is commanded by TCMS to run, the drainage valve EV (13.27) shall be commanded to be closed by TCMS at the same time with the same digital output.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.03.Zefiro-Europe.CONCEPT.478	The contact of the drainage valve (EV) shall be connected to a digital input read by TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.479	If the drainage valve (EV) remains de-energized when main compressor is ordered to be active, an event shall be generated	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.02.Zefiro-Europe.CONCEPT.HRAS003	**Parking mode** If the train is in parking mode it is necessary to use only one compressor for fill up. The primary compressor defined before the parking mode activation shall be used The compressor shall run at following pressure values: Compressor start: MR pressure (monitored on the hal train where compressor is defined active) = 650 kPa Compressor Stop = 1000 kPa It is necessary to use the same compressor and not rotate the compressor in this operating mode with following exceptions: - Compressor running more than 30 minutes (pumping time, continous pumping) (2F_05.02.Zefiro-Europe.CONCEPT.HRAS001) - Multiple compressor overheating protection triggered (as per 2F_06.03.Zefiro-Europe.CONCEPT.139)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.02.Zefiro-Europe.CONCEPT.HRAS004	**Parking mode** In parking mode, in multiple composition only 1 compressor per whole train (2 consist) shall be ordered to run The primary compressor defined before the parking mode activation and located in the consist with enabled cab shall be used	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

2F_05.02.Zefiro-Europe.CONCEPT.HRAS002	**Requirement ID empty** ~~**Parking mode**~~ ~~When compressor is in parking mode configuration TCMS will record the event "ComprParking=1" in the ambient data.~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~	~~EVENT~~	~~N/A~~
2F_05.02.Zefiro-Europe.CONCEPT.HRAS005	**Timer management entering or exiting Parking Mode:** When entering in Parking Mode, the timer counter shall be continue to run, exceeding the 3 hours without swapping the compressor. When the train status will exiting the Parking Mode the primary compressor shall swap and the timer shall reset to 0. This requirement will be higher priority to to 2F_06.03.Zefiro-Europe.CONCEPT.126, if the train switch from off to active the primary compressor is deemed to 2F_06.03.Zefiro-Europe.CONCEPT.126 if the status is changed from parking mode to normal the primary compressor is deemd to this requirement. *Note: it is necessary the compressor swap in order to avoid to use the same compressor. If the service is the same, e.g. 8 h, it will be used only one compressor if not cycled at the start after the parking.*				

ID	2F_06.03.Zefiro-Europe.TRS.48
Description	The compressor control shall equalize duty cycles of compressors in vehicle.

Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.126	The main compressor located closed to the activated cab shall be initially defined as the primary compressor within an 8 car train, if no fault related to that compressor is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.239	The main compressor located remote to the activated cab shall be initially defined as the secondary compressor within an 8 car train.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.241	When the primary compressor has run for 3 hours (running time) and the secondary compressor is available the following switch over shall be done: 1. secondary compressor shall become the primary 2. the primary compressor shall become the secondary	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.150	The run time in hours of each compressor shall be calculated and stored as condition data (accessible and resettable by maintenance staff, stored in non volatile memory by TCMS).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.152	The compressor control shall be done per 8 car unit in a multiple train (i.e. each 8 car unit in a multiple train shall be independent from each other).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.437	The BCU shall send the main pipe pressure value of sensor TCC as IP signal "Main pipe pressure TCC (T-cars)" to TCMS in kPa.	Derived Requirement	4S_06.01.-.Air Supply Systems		SIL 2
2F_06.03.Zefiro-Europe.CONCEPT.158	The BCU shall send the main pipe pressure value of sensor TCP as IP signal "Main pipe pressure TCP (DMs-cars)" to TCMS in kPa.	Derived Requirement	4S_06.01.-.Air Supply Systems		SIL 2
2F_06.03.Zefiro-Europe.CONCEPT.294	The pressure transducer TCC 13.23 read by BCU and the pressure transducer TCP 13.12 in DM-car read by BCU shall be used to read the MR pressure. The most significant valid value shall be used for compressor management, the highest value when stopping the compressor and the lowest when starting. The pressure value coming from TCC or TCP, should not be considered if the cock RICP of the relative element (car) is closed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.155	The status of all individual compressors in T4 and T5 cars shall be displayed on the IDU (Running - Not running - blocked - forced running)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

ID	2F_06.03.Zefiro-Europe.TRS.15
Description	When the pressure has decreased to limit 1 (850 kPa) the first compressor shall start.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.51	The primary main compressor shall start when MR pressure limit 1 is < 850 kPa if power is available on the 400VAC bus.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_06.03.Zefiro-Europe.TRS.52
Description	When the pressure has decreased to limit 2 (800 kPa) a second compressor shall start.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.84	The secondary compressor shall start when MR pressure limit 2 is < 800 kPa.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_06.03.Zefiro-Europe.TRS.53
Description	If the pressure decreases below limit 3 (700 kPa), a warning message shall be presented in the cab.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.87	At MR pressure limit 3 < 700 kPa an event message "Low main reservoir pressure " shall be set. It shall be an alarm message visible to the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.03.Zefiro-Europe.CONCEPT.127	Detection of low MR pressure shall be blocked during start up of a not pressurised train (<800kPa) and enabled once the pressure has reached 800 kPa at least one time.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_06.03.Zefiro-Europe.TRS.55
Description	All compressors shall stop when pressure has increased up to limit 1000 kPa.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.93	All main compressors shall stop at MR pressure >=1000 kPa.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_06.03.Zefiro-Europe.TRS.49
Description	Restart of compressors shall be prevented for at least 30 s (the time shall be adjustable).
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.165	After the compressors have been stopped the re-activation shall be possible after 60sec. The time shall be adjustable.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_06.03.Zefiro-Europe.TRS.77
Description	In case of failure also to the secondary compressor, the reinclusion of the compressor with fault to the dryer in order to terminate the service shall be possible.
Safety level	SIL 0

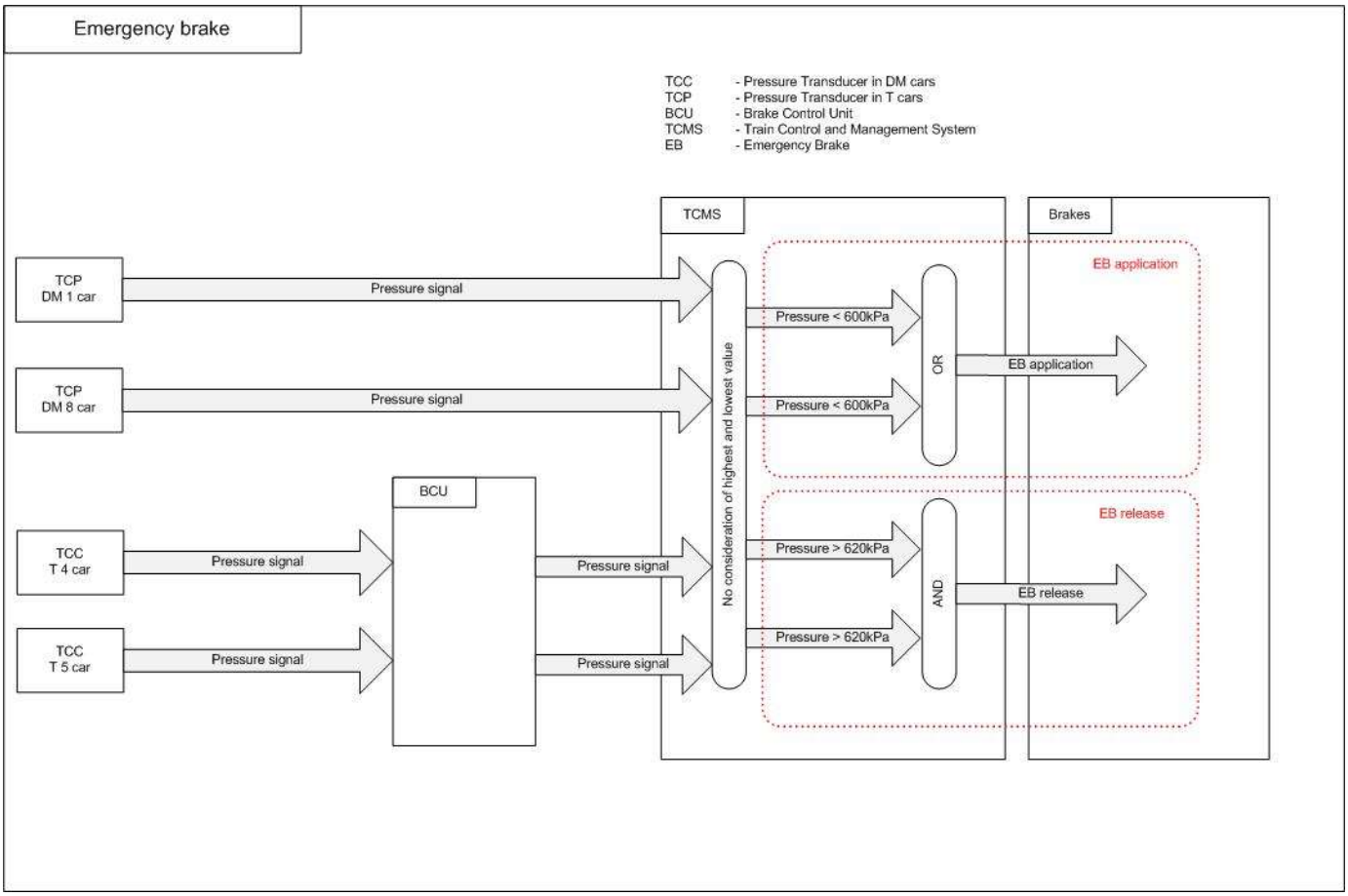
Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.510	GIVEN both compressors have been disabled AND at least one air dryer is faulty AND the working time is lower than the maximum working time THEN the driver can re-enable the not faulty compressors by “SoftKey” on IDU	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.511	GIVEN compressors not faulty have been re-enabled THEN the maximum working time of the re-enabled not faulty compressors shall be 6 hours **Note**: working time is the sum of the activation times Ton, duty-cycle % = (Ton/Toff) *100.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.512	GIVEN compressors not faulty have been re-enabled OR the working time is elapsed THEN the maintainer can reset the maximum working time of the re-enabled not faulty compressors by “SoftKey” on IDU	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.513	GIVEN not faulty compressors have been re-enabled	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

	THEN only Telediagnostic Room is informed by an event that not faulty compressors have been re-enabled on the vehicle				
2F_06.03.Zefiro-Europe.CONCEPT.514	GIVEN as long as not faulty compressors have been re-enabled (faulty air dryers bypassed) THEN driver is informed by an event that not faulty compressors have been re-enabled on the vehicle	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.515	GIVEN the working time of the re-enabled not faulty compressors has been reset by the maintainer THEN only Telediagnostic Room is informed by an event that faulty air dryers on the vehicle have been repaired	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

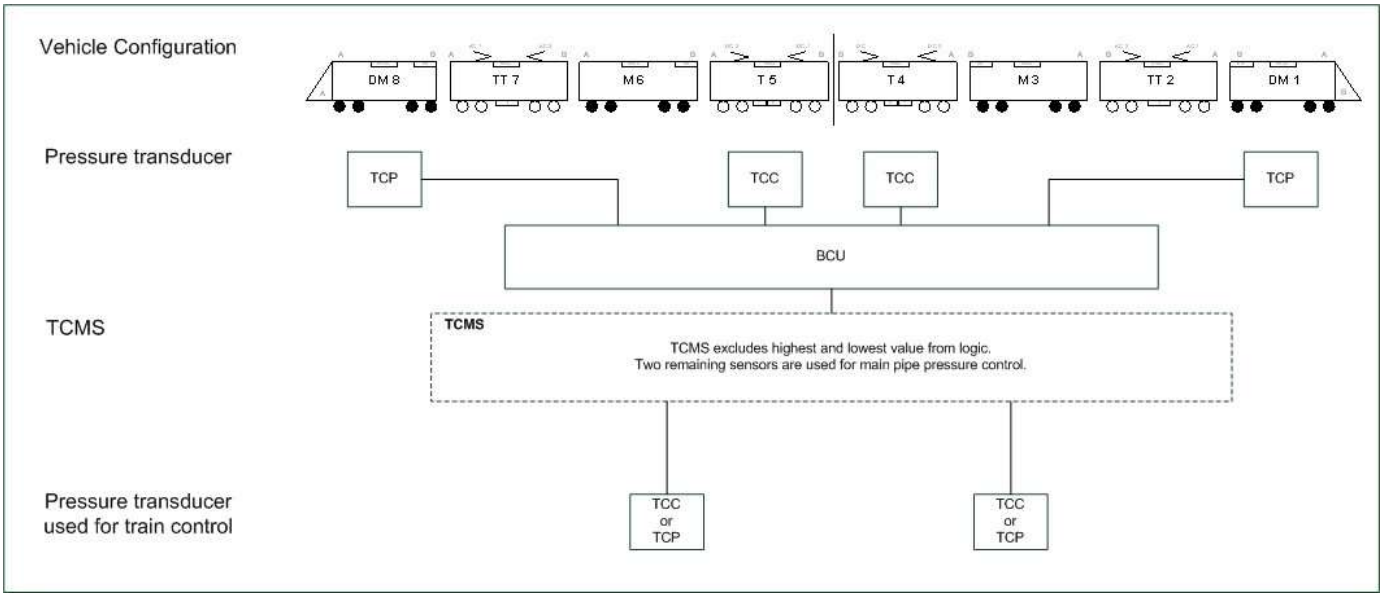
3.1.2.1 *Emergency Brake Low Main Reservoir Pressure*

ID	2F_06.03.Zefiro-Europe.TRS.54
Description	If the pressure decreases below limit 4 (600 kPa), emergency brake shall be applied.
Safety level	SIL 2

<Picture>



<Picture>



Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.465	If the pressure decreases below 650/750(open/close) kPa the pressure switch PCP (in DM1 and DM8) shall open its contacts.	Derived Requirement	4S_06.01.-.Air Supply Systems		SIL 2
2F_06.03.Zefiro-Europe.CONCEPT.466	The pressure switch PCP shall close its contacts at pressure > 750 kPa.	Derived Requirement	4S_06.01.-.Air Supply Systems		SIL 2
2F_06.03.Zefiro-Europe.CONCEPT.90	Emergency brake request shall be sent out if main pipe pressure is below 600kPa.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_06.03.Zefiro-Europe.CONCEPT.234	Emergency brake shall be released if main pipe pressure is above 620 kPa.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_06.03.Zefiro-Europe.CONCEPT.406	The following sensors shall be used safe main pipe pressure supervision: 1. the pressure transducers in DM-cars (TCP) 2. the pressure transducers in T-cars (TCC) If one pressure transducer is not valid it shall be excluded from the logic to calculate the main pressure.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_06.03.Zefiro-Europe.CONCEPT.436	The main pipe pressure value shall be calculated based on valid sensors: 1. 4 pressure sensors are valid: TCMS shall exclude the highest and lowest value from consideration, TCMS shall take the lowest pressure value of the remaining two sensors 2. 3 pressure sensors are valid: TCMS shall exclude the highest and lowest value from consideration, TCMS shall take the remaining sensor 3. 2 pressure sensors are valid: TCMS shall take the lowest pressure value of the two remaining sensors 4. only 1 pressure sensors is valid: emergency brake shall be applied	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_06.03.Zefiro-Europe.CONCEPT.128	When emergency brake is applied due to low pressure an event "Emergency brake applied due to low main pipe pressure" shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.2 Monitoring of Main Compressor

3.2.1 Function Design

3.2.1.1 *Normal Condition*

Compressors are cooled and normally not overheated.
When train is started compressor will run longer time because MRP system is empty.
In normal running conditions one compressor is enough to fill up MRP system. Air supply system is redundant in case of compressor failure.

3.2.1.2 *Failure Cases*

1. Compressor is overheated
2. Compressor is running without an order
3. Compressor is not running on order
4. Contactors for compressors are not activated
5. Too high main reservoir pressure

3.2.1.3 *Failure Detection*

1. An event is generated and compressor is isolated if compressor is overheated the third time
2. An event is generated
3. An event is generated
4. An event is generated
5. An event is generated

3.2.1.4 *Failure Handling*

1. Compressor shall be stopped and secondary compressor shall run as primary compressor. Overheated compressor is allowed to start again when it is cooled down. If compressor is overheated 3 times in 24 hour it will be stopped permanently until a reset is performed by maintenance staff. No effect on train operation.
2. Compressor is working against safety valve 13.13 when MR system is overcharged. No effect on train operation.
3. Secondary compressor will be start-up, no impact on train operation.
4. Compressor is not started, secondary compressor will be used. No impact on train operation.
5. Compressor is working against safety valve 13.13 when MR system is overcharged, no impact on train operation.

3.2.2 Requirements

ID	2F_06.03.Zefiro-Europe.TRS.19
Description	The TCMS shall monitor the main compressor to prevent damage due to high temperature and long running time.
Safety level	SIL 0

Main compressor abnormal running conditions

1. Overheating of motor in main compressor.
   Main compressors are protected against overheating by thermal protection which measures the compressor motor temperature. If main compressor is overheated the secondary compressor will be used as primary compressor. The runtime timer will reset as soon as the overheated compressor is cooled down. It can be run as secondary compressor. Event will be reset automatically by the system before the overheated compressor can start again, or manually by maintainer master reset if the overheat occurred three times.
2. The main compressor contactor is not connecting when start request is sent. Compressors will not start
3. The main compressor contactor is not disconnecting when stop request is sent. Safety valve in main line will be activated. Compressors will be overheated.
4. If a compressor is continuously running for 30 minutes (with 2 compressors healthy) it shall necessary to switch compressor master to slave in 30 minutes instead of 180 minutes.

5. During parking mode it is necessary to run 1 compressor per train (1 per single unit or 1 per multiple unit). Pressure threshold is reduced to 6,5 bar in order to limit the number of compressor starts in this condition and warm up the compressor in a low duty cycle mode. If the parking mode is not active the compressor management switch to normal.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.501	Electric motors in main compressors shall be diagnostized regarding overheating. The thermal compressor switch shall be connected to TCMS digital input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.56	Electric motors in main compressors shall be diagnostized regarding overheating in windings by thermal switch and monitored when compressors are running.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.135	The compressor contactor shall be supervised and when a start or stop request is sent and not confirmed reading the contactor position, an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.477	The compressor contact shall be connected to a digital input read by TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.136	If main compressor is overheated indicated by thermo switch PTC in main compressor an event "Main Compressor temperature too high." shall be generated. The event shall be reset automatically after 30 min after up to two overheating events.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.138	Main compressors are allowed to be stopped 2 times because of high temperature. After 3rd time of start-up with high temperature an event "Start of compressor is blocked due to overheating." shall be set and the compressor shall be blocked and not be allowed to start again until a reset is performed by maintenance staff.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.137	If main compressor is overheated then the start signal for compressor shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.285	The overheated main compressor can be started again if the overheat failure event is no longer active and compressor has been cooled down for 30 minutes.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.246	Maintenance personnel shall be able to reset a blocked compressor from IDU after rectification of a permanent error like overheating.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.139	If the primary main compressor is stopped due to high temperature then the secondary compressor shall work as primary main compressor.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.491	If one of the main compressors exceeds 30 starts per hour, it shall be blocked for 30min.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.140	An event "Main compressor running when not ordered." shall be generated if the compressor running signal is high without order to start compressor.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.142	When both compressors are not faulty and any of them has run more than 30 minutes a long run time running event	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

	"Main compressor has run more than 30min continuously" shall be set.				
2F_06.03.Zefiro-Europe.CONCEPT.143	During start up of the not pressurised train (< 800 kPa) or with only one compressor available the run time shall be 60 minutes before an event "Main compressor too long running time at start up, more than 60 min continuously." due to long run time is set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.02.Zefiro-Europe.CONCEPT.HRAS001	If it is passed 60 minutes from the start up and event 2F_06.03.Zefiro-Europe.CONCEPT.142 and there are 2 compressors healty in the single unit it is necessary to set the compressor referred to the event as secondary and set the timer 2F_06.03.Zefiro-Europe.CONCEPT.241 to 0 min (the timer is reduced). This requirement is per single unit, in multiple unit each unit shall be considered separately. The compressor set as secondary is not permanently secondary. When it is expired the normal cycle time TCMS will try to set this compressor as primary. If the problem is restored it will be kept primary if the event 2F_05.02.Zefiro-Europe.CONCEPT.HRAS001 will be set again it will be switched as secondary again after 30 min of continous pumping.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.144	An event shall be set if the main reservoir pressure is >1030kPa for more than 2 seconds.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_06.03.Zefiro-Europe.TRS.51
Description	If Tilt system is in-active and the air consumption is high an event shall be generated.
Safety level	SIL 0

ID	2F_06.03.Zefiro-Europe.TRS.50
Description	Leakages in the air supply system shall be detected if exceeding defined value.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.111	If MR pressure is > 700 kPa and brake pipe pressure is > 480 kPa the pressure decrease shall be read. If the pressure decrease > 100 kPa/min a leakage event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.03.Zefiro-Europe.CONCEPT.447	If MR pressure is > 700 kPa and brake pipe pressure is > 480 kPa the pressure decrease shall be read. If the pressure decrease > 200 kPa/min an event shall be set to indicate major leakage.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.3 Forcing of Main compressor

ID	2F_06.03.Zefiro-Europe.TRS.69
Description	It shall be possible to manually from the cab to force the running of the main compressors.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.386	There shall be a pushbutton on the IDU with toggling function for force start of the main	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

	compressor. It shall be possible to start each individual compressor in the train (also multiple train). The main compressors are located in T-cars.				
2F_06.03.Zefiro-Europe.CONCEPT.393	The selected compressor shall start if the button is pressed. The compressor shall stop if the button is pressed again or the compressor running time is > 180s . When the force push button is toggled back or time limit is reached the compressor shall go back to automatic compressor control.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.384	The compressors shall be able to be forced on/off by TCMS SW parameter setting.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.4 Monitoring of Main Pipe

3.4.1 Function Design

3.4.1.1 *Normal condition*

The main pipe pressure value is in normal condition between 850kPa and 1000kPa.
All sensors are valid and do not indicate a pressure that differs from an average value with a tolerance of +/- 50kPa.
The isolation cock RISP 13.22 is not isolated and not faulty.

3.4.1.2 *Failure Cases*

1. Faulty pressure sensor
2. Isolation cock RISP 13.22 status is non-isolated in isolated position
3. Isolation cock RISP 13.22 status is isolated in non-isolated position

3.4.1.3 *Failure detection*

1. Faulty pressure sensors will be indicated by events
2. Reduced filling time. Safety valve 13.13 releases pressurized air.
3. Isolated cock will be indicated in IDU. Visual check will show not isolated cock.

3.4.1.4 *Failure handling*

1. Remaining pressure sensors are used for main pipe monitoring. No impact on train operation.
2. Second compressor is active in parallel. No impact on train operation.
3. One compressor is enough to fulfil the need of compressed air in the train. No effect on train operation.

3.4.2 Requirements

ID	2F_06.03.Zefiro-Europe.TRS.62
Description	MR pipe shall be diagnostized when train is in operation.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.214	When the BCU send the signal "DIA TCP transducer faulty" in DM1 car, an event "Pressure transducer faulty in DM1-car" shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.215	When the BCU send the signal "DIA TCP transducer faulty" in DM8 car, an event "Pressure transducer faulty in DM8-car" shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.199	When the pressure transducer TCC 13.23 in T4 car is out of range the BCU shall sent a signal over IP to TCMS.	Derived Requirement	4S_06.01.-.Air Supply Systems		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.212	When the BCU send the signal "DIA TCC main pipe pressure transducer faulty" in T4 car, an event "Pressure transducer faulty in T4-car" shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_06.03.Zefiro-Europe.CONCEPT.200	When the pressure transducer TCC 13.23 in T5 car is out of range the BCU shall sent a signal over IP to TCMS.	Derived Requirement	4S_06.01.-.Air Supply Systems		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.213	When the BCU send the signal "DIA TCC main pipe pressure transducer faulty" in T5 car, an event "Pressure transducer faulty in T5-car" shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.197	A pressure transducer shall not be used for compressor control due to the following reasons: 1. Faulty according to the evaluation matrix CONCEPT.464between pressure transducers and pressure switches 2. Valid signal from BCU not set 3. Pressure value out of range 4. Communication error with BCU 5. Isolation of related car by RICP cock where pressure sensor is located	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.286	A pressure transducer that is faulty shall not be used for compressor control.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.159	The fault messages according to the table below shall be generated in case the signals from the pressure indicators differ, providing no communication errors with the brake unit are detected.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.464	DM8 car / DM1 car TCP in kPa | PCP high > 750kPa low < 650KPa | TCP in kPa | PCP high > 750kPa low < 650KPa | Fault indication > 780 | high | > 780 | high | ok > 780 | high | > 780 | low | faulty PCP in DM1 > 780 | low | > 780 | high | faulty PCP in DM8 < 620 | high | > 780 | high | faulty TCP in DM8 > 780 | high | < 620 | high | faulty TCP in DM1 < 620 | low | < 620 | low | ok < 620 | low | < 620 | high | faulty PCP in DM1 < 620 | high | < 620 | low | faulty PCP in DM8 > 780 | low | < 620 | low | faulty TCP in DM8 < 620 | low | > 780 | low | faulty TCP in DM1	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.202	There shall be a fault message generated in case the signals from the pressure sensors in T cars (TCC) differ according to average pressure value of all valid pressure sensors (TCP and TCC). The allowed deviation from the average value is +/- 100kPa.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.163	The value of all transducers and switches shall be indicated individually on the IDU. DM cars: 1. MR transducer TCP (value 0...1100kPa) 2. MR pressure switch 650/750kPa PCP (open/closed) T cars: 1. MR transducer TCC (value 0...1100kPa)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.377	When air supply cut out cock RIHVAC (20.03/01;20.03/18) in DM car is closed an event "Air supply to HVAC in DM x is closed" shall be generated for the specific DM car where the cock is closed.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.379	When air supply cut out cock RIHVAC (20.03/02;20.03/17) is closed an event "Air supply to HVAC 1 in DM x for saloon is closed."	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

	shall be generated for the specific car where the cock is closed.				
2F_06.03.Zefiro-Europe.CONCEPT.380	When air supply cut out cock RIHVAC (20.03/03;20.03/16) is closed an event "Air supply to HVAC 2 in DM x for saloon is closed." shall be generated for the specific car where the cock is closed.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.401	If a compressor is isolated from MR system by cock RISP 13.22/01 it shall be blocked and therefore considered as not available. An event "Compressor isolated from MR system" shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.5 Air Dryer

3.5.1 Function Design

T4- and T5-car contain a two tower air dryer 13.05 which functionality is supervised by Air dryer control unit. Faults are indicated to TCMS by one signal line.

3.5.1.1 *Normal Condition*

During 120 s interval the actual tower is working as dryer for 120 s and it is dried for 120 s by air.
In normal condition air towers are switching in 120 s intervals. During this interval one of the towers is working as dryer for compressed air and the other tower is dried and blown clean of water with compressed air.
Air dryers are used even when external air supply is connected to dry workshop air before it enters to trains MRP system.

3.5.1.2 *Failure Cases*

1. Towers are not switching in 120s intervals
2. Towers are switching when compressors are not running or external air supply is not connected.

3.5.1.3 *Failure Detection*

1. Event shall be generated
2. Event shall be generated

3.5.1.4 *Failure Handling*

1. Humid air will pass to MR system, damage possible. Affected compressor will be deactivated, no impact on train operation.
2. No damages but switching of air towers is not necessary. No impact on train operation.

3.5.2 Requirements

ID	2F_06.03.Zefiro-Europe.TRS.24
Description	The TCMS system shall monitor that the air dryer switches between the two drying towers.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.134	The air dryer shall run when corresponding compressor is running or external air supply is connected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.219	Air towers shall be switched every 120 s if they are active (compressor running or external air connected).	Derived Requirement	4S_06.01.-.Air Supply Systems		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.61	if air dryer is reported as faulty, a fault message shall be generated and the local compressor shall be disabled.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.130	The air dryer diagnostic output in the main air supply unit shall only be supervised when compressor is running or external air supply is opened (RIPO cock isolated)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_06.03.Zefiro-Europe.CONCEPT.133	The switching intervals of the towers shall ensure that no humid air can enter main reservoir system. The quality of the compressed air shall comply with purity classes 2 2 NA (output of air supply system) .	Derived Requirement	4S_06.01.-.Air Supply Systems		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.502	Air dryer start signal shall be connected to TCMS output. The isolation cock RIPO in isolated position shall start the air dryer. The air dryer diagnostic output shall be connected to TCMS input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

3.6 External Air Supply

3.6.1 Function Design

3.6.1.1 *Normal Condition*

When external air supply is connected, workshop air is used to fill MR system with air. The air connection is in the T cars. Air is connected if the cock RIPO is isolated (i.e. open), this has to be done manually. It is necessary to have the Low Voltage active and the powering to the air dryer to feed the pneumatic line.

3.6.1.2 *Failure Cases*

When external air supply is connected, the cock status or the connection of the external air hose is not indicated correctly:

1. The isolation cock RIPO status indicates "isolated" in non isolated position
2. The cock status indicates "not isolated" in isolated position

3.6.1.3 *Failure Detection*

1. External air connected but not indicated
2. External air indicated but no connection

3.6.1.4 *Failure Handling*

1. No error handling possible.
   Driver is not informed that hoses are connected to train, driver can drive the train with external hose connected. Visual check has to be performed before starting the operation.
2. Traction block is activated while no external air supply is connected, cock indication needs to be overridden by driver as cut out on the IDU. No impact on train operation when overridden.

3.6.2 Requirements

ID	2F_06.03.Zefiro-Europe.TRS.29
Description	When external air supply is connected: 1. all compressors shall be disabled 2. traction shall be blocked
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.66	If any external air supply isolation cock switch (RIPO 13.29 in T cars) indicates that external air supply is connected (RIPO=isolated) all compressors shall be disabled regardless of starting request comes from Pressure sensor or Force button on IDU (also at multiple train configurations).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.385	The driver shall have the possibility to override the effect of the RIPO switches to the external air supply connection. Each switch shall be able to be overriden individually from the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

2F_06.03.Zefiro-Europe.CONCEPT.405	The override of the external air connected switches shall be a toggle function from the IDU button. It shall be deactivated if the button is pressed again.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.156	Traction block shall be activated (can be overridden from IDU) if external air supply is connected in any car, i.e. the RIPO switch in any T car indicates "isolated" (external supply open).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.157	When external air supply is connected, the status and location shall be shown on the IDU: T4 or T5 car. Following status of the RIPO isolation cock shall be shown on IDU: Isolation (External air supply possible, cock open) Not-isolated (cock closed) Overridden	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.257	An event shall be generated if the RIPO cock is isolated (i.e. open), indicating "External air supply is connected"	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.7 Control of Auxiliary Compressor

3.7.1 Function Design

3.7.1.1 *Normal Condition*

Auxiliary compressor normal running situation
Normally one auxiliary compressor is enough to lift up the pantograph (only one pantograph needed). The other auxiliary compressor is a back up. The auxiliary compressor in the half train where is located the pantograph to lift is started to fill pantograph system when order is given.

The compressor supply the pantographs with pressurized air to be able to get high voltage when the main reservoirs pressure is low < 600 kPa.
Auxiliary compressors are placed in T4- and T5-car together with main compressors and supplied by 110 VDC from battery.

One auxiliary air compressor supplies both pantographs in a local 4 car unit.

A bypass function allows to raise the pantograph in case of TCMS unavailability. The activation can be done locally on the respective car (T4 or T5)

The following components are used:
1. Pressure switch PSP 08.09 on the auxiliary reservoir in T4/5 cars.

3.7.1.2 *Failure Cases*

1. Faulty pressure switch
2. Auxiliary compressor is overheated
3. Long running time
4. Auxiliary compressor is running without an order or when pressure in pantograph circuit is >700 kPa.
5. The auxiliary compressor contactor is not connecting when start request is sent.
6. The auxiliary compressor contactor is not disconnecting when stop request is sent.
7. high leakage in one (local) auxiliary air system

3.7.1.3 *Failure Detection*

1. An event shall be generated if one of auxiliary compressors does not start when start order is valid.
2. An event shall be generated
3. An event shall be generated
4. An event shall be generated
5. An event shall be generated
6. An event shall be generated
7. An event for long run time will be generated.

3.7.1.4 *Failure Handling*

1. The other auxiliary compressor can be used. The Aux compressor can also be forced running, no impact on train operation.
2. Overheated auxiliary compressor shall be stopped when it is overheated, no impact on train operation.
   The second auxiliary compressor can be used to fill pantograph system in second 4 car unit.
3. Auxiliary compressor shall be stopped if running time is too long. Other compressor can be used, no impact on train operation.
4. Auxiliary compressor will run against safety valve 850 kPa and will be overheated and disabled, no impact on train operation.
5. The second auxiliary compressor can be used, no impact on train operation.
6. Safety valve in auxiliary compressor system will be activated. Auxiliary compressor will be overheated, no impact on train operation.
7. The second auxiliary compressor can be used. No pantograph can be lifted in defective 4 car unit, train operation can be finished until end of operational day.

3.7.2 Requirements

ID	2F_06.03.Zefiro-Europe.TRS.35
Description	The auxiliary compressor shall be started: 1. depending on which pantograph is requested to raise AND 2. main reservoir pressure too low for lifting the pantograph
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.71	When the pantograph is ordered to be raised and the auxiliary reservoir pressure is low (pressure switch PSP) the auxiliary compressor installed on the same half train as the pantograph that is ordered to be raised shall be started.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.169	The auxiliary compressor shall be stopped if the pantograph is ordered to be lowered or if the auxiliary reservoir pressure is high (pressure switch PSP).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.383	The maximum run time for the auxiliary compressor shall be 10 min. The time shall be adjustable by parameter setting.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.296	If auxiliary compressor has run 10 min an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.03.Zefiro-Europe.CONCEPT.422	If thermo switch indicate aux. compressor overheated the number of overheating occurrences shall be counted by TCMS. When the compressor has been overheated three times during 1 hour the compressor shall be blocked until a maintainer reset is performed. Note: The aux. compressor will be stopped by hardware when an over temperature occurs.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.423	If thermo switch PTC is indicating aux. compressor overheated an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.03.Zefiro-Europe.CONCEPT.490	If the auxiliary compressor exceeds 10 starts per hour it shall be blocked for 30min.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_06.03.Zefiro-Europe.CONCEPT.505	It shall be possible to cut-out each auxiliary compressor by IDU. When an aux. compressor is cut-out a diagnostic event will be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_06.03.Zefiro-Europe.CONCEPT.220	It shall be indicated on the IDU the status of the auxiliary compressor. 1. Aux compressor running/not running/blocked/cut out T4 2. Aux compressor running/not running/blocked/cut out T5	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_06.03.Zefiro-Europe.CONCEPT.175	The auxiliary compressors contactors shall be supervised and when a start request is	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

	sent and not confirmed with auxiliary compressors running status, an event shall be generated.				
2F_06.03.Zefiro-Europe.CONCEPT.476	The compressor contactor, thermal switch PTC and the pressure switch PSP shall be connected to digital inputs read by TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_06.03.Zefiro-Europe.CONCEPT.173	An event shall be set if the auxiliary compressors running signal is high without order indicating that the compressor is running without order or that the manual bypass is activated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.03.Zefiro-Europe.CONCEPT.400	If the PSP indicates low pressure when the main reservoir pressure is above 850 kPa then an event shall be set indicating that the PSP switch is faulty.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_06.03.Zefiro-Europe.TRS.42
Description	The auxiliary compressor shall stop if the main compressors have taken over the air supply (i.e. 3~ AC and Main Reservoir Pressure available).
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.96	When main compressors are running and main reservoir pressure is > 600 kPa the auxiliary compressors shall be disabled.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_06.03.Zefiro-Europe.TRS.41
Description	The aux compressor shall run if manually requested by push button in cab.
Safety level	N/A

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.99	There shall be a button on the IDU to toggle the auxiliary compressor on/off. One button for the compressor in T4 and one button for T5 shall be provided. The status on/off shall be indicated.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_06.03.Zefiro-Europe.CONCEPT.171	Auxiliary compressors shall start if the push button on the IDU is pressed and the auxiliary compressor is not running.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.238	If it has been manually started the auxiliary compressor shall stop: 1. if the button on the IDU is pressed again or 2. if it has been running for more than 10 min.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_06.03.Zefiro-Europe.CONCEPT.272	An event shall be set if the auxiliary compressor is manually operated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.7.2.1 *Bypass function for Auxiliary compressor*

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.494	There shall be a control device to allow the auxiliary compressor to run without TCMS for each auxiliary compressor. The device shall be located close to the respective auxiliary compressor.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

ID	2F_06.03.Zefiro-Europe.TRS.40
Description	The aux compressor shall be disabled to run if the max run time is exceeded.
Safety level	N/A

Long running time is an indication of leakage in pantograph system or faulty compressor.

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_06.03.Zefiro-Europe.CONCEPT.102	In case the compressor runs for longer time than 10 minutes, an event "Auxiliary compressor has run more than 10 min continuously" shall be generated and this auxiliary compressor shall be disabled for 60 min (adjustable by parameter).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.03.Zefiro-Europe.CONCEPT.172	The run time of the auxiliary compressors shall be monitored and stored as condition data.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.8 Suspension Supervision

3.8.1 Function Design

3.8.1.1 *Normal Condition*

This part is to supervise cut-out cocks for secondary suspension. Normally these cut-out cocks are open. If they are closed there is a reason for that, normally a suspension bellow which has been cracked.
There is one cock for the secondary suspension in each car: RISOSP
There is one pressure sensor for each bogie: TS1...TS8

3.8.1.2 *Failure Cases*

1. Secondary suspension is isolated in car (RISOSP 05.02 isolated)
2. Pressure sensor TS fault (supply cut) or out of range
3. Pressure sensor is indicating wrong pressure value

3.8.1.3 *Failure Detection*

1. Event shall be set
2. Event shall be set
3. No detection possible

3.8.1.4 *Failure Handling*

1. Train speed shall be limited if secondary suspension is isolated
2. Train speed shall be limited if secondary suspension status is not known, sensor shall be checked by maintaince
3. Train speed shall be limited, no detection possible

3.8.2 Requirements

ID	2F_06.03.Zefiro-Europe.TRS.39
Description	The air suspension of bogies shall be supervised and fault events shall be reported via the diagnostic system.
Safety level	N/A

The air suspension supervision of bogies regarding pressure sensors shall be supervised according to SFRS brake system.

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.184	The supervision of the cut-out cock RISOSP is done according to SFRS Brake Control chapter 3.1.2.2.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_06.03.Zefiro-Europe.TRS.68
Description	If the secondary suspension is not working, the train speed shall be limited. Note: The result of safety analysis from Bogie may increase the SIL to SIL 2.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL

2F_06.03.Zefiro-Europe.CONCEPT.389	Fault "Secondary Suspension faulty" shall be indicated to the driver if any of the following conditions is true: 1. any pressure sensor TS is faulty in train 2. any pressure value of sensor TS is below < 200kPa	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.498	If a fault "secondary suspension faulty" is active: 1. Train speed shall be limited by TCMS to 230km/h 2. The speed limit shall be displayed in IDU	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.9 Provision for Tilt system

3.9.1 Function Design

Tilting function is controlled by increasing/decreasing pressure in air bellows for secondary suspension system.
The tilting system requires more compressed air than ordinary suspension system that is the reason for different threshold values for compressor start/stop.
(Compressor control pressure sensor TCC in T4/T5. MR -pressure control TCP in DM1/DM8)
The tilting system is not present on the ETR1000 VZI R1 trains.

3.9.1.1 *Normal Condition*

Tilting system is normally in operation when train speed is > 300 km/h. MRP pressure shall be between 800 - 1000 kPa when tilting is working to guarantee the compressed air delivery to air bellows.

3.9.1.2 *Failure Cases*

The failure cases, detection and handling for tilting will be defined and described in the relevant SFRS for Tilting system.

3.9.2 Requirements

ID	2F_06.03.Zefiro-Europe.TRS.45
Description	When the pressure has decreased to limit 1 (900 kPa) the first compressor shall start. When the pressure has decreased to limit 2 (850 kPa) a second compressor shall start. Compressors shall stop when pressure has increased up to limit 1000 kPa.
Safety level	SIL 0

Nr	Description	Applicability Details	Derived to	User Interface	SIL
2F_06.03.Zefiro-Europe.CONCEPT.81	The primary main compressor shall start when MR pressure< 900 kPa and if the tilting function is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.176	The secondary compressor shall start when MR pressure is < 850kPa and if the tilting function is active.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.177	The primary and secondary compressor shall stop when the MR pressure > = 1000 kPa.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.231	Main compressors shall be started when triggered by the tilting control system. This will be done when entering curves.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.276	If tilting is active (as indicated by the tilting system) and the Main compressor indicates a running time of more than 20 min, a signal shall be sent to the tilting system to demand the deactivation of the tilting function.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.03.Zefiro-Europe.CONCEPT.461	If tilting is active (as indicated by the tilting system) and the Main compressor indicates a running time of more than 20 min, an event "long running time while tilting" shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_06.03.Zefiro-Europe.CONCEPT.277	If the cock RISOSP for secondary suspension is closed in any car, a signal shall be sent to the	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	tilting system to demand the deactivation of the tilting function.				

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_06.03.Zefiro-Europe.CONCEPT.295	Auxiliary compressor pressure switch PSP (08.09)	L	2F_06.03.Zefiro-Europe.CONCEPT.71
2F_06.03.Zefiro-Europe.CONCEPT.457	Auxiliary compressor running (contactor closed)	L	2F_06.03.Zefiro-Europe.CONCEPT.102 2F_06.03.Zefiro-Europe.CONCEPT.175 2F_06.03.Zefiro-Europe.CONCEPT.476
2F_06.03.Zefiro-Europe.CONCEPT.392	Auxiliary compressor temperature switch PTC	L	2F_06.03.Zefiro-Europe.CONCEPT.422 2F_06.03.Zefiro-Europe.CONCEPT.423
2F_06.03.Zefiro-Europe.CONCEPT.297	Air dryer 1 active (pressure switch PTA)	L	2F_06.03.Zefiro-Europe.CONCEPT.61
2F_06.03.Zefiro-Europe.CONCEPT.298	Air dryer 2 active (pressure switch PTB)	L	2F_06.03.Zefiro-Europe.CONCEPT.61
2F_06.03.Zefiro-Europe.CONCEPT.299	Air supply cock RIHVAC in DM, HVAC cab unit isolated	L	2F_06.03.Zefiro-Europe.CONCEPT.377
2F_06.03.Zefiro-Europe.CONCEPT.300	Air supply cock saloon, all cars, HVAC unit 1 isolated	L	2F_06.03.Zefiro-Europe.CONCEPT.379
2F_06.03.Zefiro-Europe.CONCEPT.301	Air supply cock saloon, all cars, HVAC unit 2 isolated	L	2F_06.03.Zefiro-Europe.CONCEPT.380
2F_06.03.Zefiro-Europe.CONCEPT.278	Main compressor running (contactor is closed).	L	2F_06.03.Zefiro-Europe.CONCEPT.155
2F_06.03.Zefiro-Europe.CONCEPT.480	Drainage valve EV	L	2F_06.03.Zefiro-Europe.CONCEPT.479
2F_06.03.Zefiro-Europe.CONCEPT.440	Main compressor temperature sensor PTC	L	2F_06.03.Zefiro-Europe.CONCEPT.136
2F_06.03.Zefiro-Europe.CONCEPT.441	External air supply isolation cock RIPO (13.29)	L	2F_06.03.Zefiro-Europe.CONCEPT.66
2F_06.03.Zefiro-Europe.CONCEPT.442	Isolation cock main air supply unit RISP (13.22)	L	2F_06.03.Zefiro-Europe.CONCEPT.401
2F_06.03.Zefiro-Europe.CONCEPT.467	Main pipe pressure switch 600kPa (PCP) in DM car	L	2F_06.03.Zefiro-Europe.CONCEPT.464
2F_06.03.Zefiro-Europe.CONCEPT.468	RISOSP isolated	L	2F_06.03.Zefiro-Europe.CONCEPT.389
2F_05.02.Zefiro-Europe.CONCEPT.HRAS002	ComprParking		2F_05.02.Zefiro-Europe.CONCEPT.HRAS002

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_06.03.Zefiro-Europe.CONCEPT.303	Auxiliary compressor contactor close	L	2F_06.03.Zefiro-Europe.CONCEPT.175
2F_06.03.Zefiro-Europe.CONCEPT.302	Main compressor contactor close	L	2F_06.03.Zefiro-Europe.CONCEPT.46 2F_06.03.Zefiro-Europe.CONCEPT.135

Command air dryer on

4.2 Bus Interface

Brake control ICD (EC130019101B) is defining the signal interface also for the air supply system.

Nr	Functional name	Type	Linked to requirement
2F_06.03.Zefiro-Europe.CONCEPT.304	Main pipe pressure sensor TCC (T-cars) in kPa (Scaling 10 = 1kPa).	A	2F_06.03.Zefiro-Europe.CONCEPT.437
2F_06.03.Zefiro-Europe.CONCEPT.472	Main pipe pressure sensor TCC in T4/T5 car faulty	L	2F_06.03.Zefiro-Europe.CONCEPT.199
2F_06.03.Zefiro-Europe.CONCEPT.473	Main pipe pressure sensor TCP in DM1/DM8 car faulty	L	2F_06.03.Zefiro-Europe.CONCEPT.200
2F_06.03.Zefiro-Europe.CONCEPT.458	Isolation cock for secondary suspension (RISOSP) open	L	2F_06.03.Zefiro-Europe.CONCEPT.184 2F_06.03.Zefiro-Europe.CONCEPT.277
2F_06.03.Zefiro-Europe.CONCEPT.459	Pressure sensor TS1 for secondary supsension (bogie 1)	A	2F_06.03.Zefiro-Europe.CONCEPT.389
2F_06.03.Zefiro-Europe.CONCEPT.460	Pressure sensor TS2 for secondary supsension (bogie 2)	A	2F_06.03.Zefiro-Europe.CONCEPT.389
2F_06.03.Zefiro-Europe.CONCEPT.492	Main pipe pressure sensor TCP (DM-cars) in kPa (Scaling 10 = 1kPa).	A	2F_06.03.Zefiro-Europe.CONCEPT.158

5 References

SFRS Brake - FA020023100
SyRS Brake System - EC010019210
ICD Brake Control - EC130019101B